Matrix inequalities and norms

Product inequality for disjoint principal minors of Wishart matrices

Difficulty: challenging **Importance:** interesting to specialist

Status: Partially resolved

Topic: Random Gram determinants and Gaussian product inequalities

Rating rationale: The two-block result does not yield the general joint inequality, which couples arbitrarily many determinant powers. The main audience studies random Gram matrices, covariance determinants, and Gaussian volume inequalities.

Statement

Let $d, p_1, \dots, p_d$ be positive integers, set $p = \sum_{i=1}^{d} p_i$, and choose a real number $\alpha > p - 1$ and a real symmetric positive definite matrix $\Sigma \in \mathbb{R}^{p \times p}$.

Let $X \sim W_p(\alpha, \Sigma)$ mean that X has density proportional to

$$(\det X)^{(\alpha - p - 1)/2} \exp\!\left[-\tfrac{1}{2}\operatorname{tr}(\Sigma^{-1} X)\right]$$

on the cone of real symmetric positive definite matrices, with respect to Lebesgue measure on their independent entries. Partition $X = (X_{ij})_{i,j=1}^{d}$ so that X_{ij} has size $p_i \times p_j$.

Conjecture (Genest–Ouimet–Richards). For every such choice and every $\nu_1, \dots, \nu_d \geq 0$,

$$\mathbb{E}\left[\prod_{i=1}^{d} (\det X_{ii})^{\nu_i}\right] \geq \prod_{i=1}^{d} \mathbb{E}\left[(\det X_{ii})^{\nu_i}\right].$$

All displayed expectations are finite. Degrees of freedom and exponents need not be integers. All block counts and block sizes belong to this one conjecture.

Known cases and numerical significance

The conjecture is proved for $d = 2$. Block-diagonal Σ makes the diagonal blocks independent and gives equality. Results for negative determinant powers address a different exponent regime.

For integer $\alpha \geq p$, one may write $X = Y^{\mathsf{T}} Y$, where Y has α independent Gaussian rows with covariance Σ. Then $\det X_{ii}$ is the squared volume generated by the columns in block i. The problem compares joint volume moments with the corresponding product of marginal moments, linking it to random Gram matrices and volume methods.

References and status check

- C. Genest, F. Ouimet and D. Richards, *On the Gaussian product inequality conjecture for disjoint principal minors of Wishart random matrices*, Electronic Journal of Probability 29 (2024), paper 166. DOI; current arXiv v3, dated 2024-10-01. Conjecture 1.1, equation (6), states the target; Definition 2.4 defines the distribution; Remarks 1.2–1.3 record the two-block case and moment finiteness.
- C. Genest, F. Ouimet and D. Richards, *An explicit Wishart moment formula for the product of two disjoint principal minors*, Proceedings of the AMS 153 (2025), 1299–1311. DOI; full text. This proves the two-block case.
- R. E. Gaunt and F. Ouimet, *On Wilks' problem: Exact recursive formulas via Stein's method for the joint moments of disjoint principal minors of Wishart random matrices*, arXiv:2607.03984 (2026). Its moment recursions do not assert the displayed inequality for all real exponents.
- F. Ouimet and D. Greaves, *A proof of the strong Gaussian product inequality conjecture* (2026), author-posted manuscript, Theorem 2.1. This claims a scalar Gaussian-coordinate product inequality; the introduction separately identifies Wishart extensions. Its stated result does not cover arbitrary determinant blocks. This entry does not assess that manuscript's proof.

On 2026-09-11, checked the source's current version, these later papers and Ouimet's publication list, and searched the exact title, Wishart product inequalities, Conjecture 1.1, and 2025–2026 proof/counterexample combinations. No full resolution of the displayed determinant-block target was found. This is a bounded status check.